

Application of Flowform® Technology in the Sewage Water Treatment Plant Soerendonk (NL)

Situation and Ambition

Hidden behind thick vegetation, the SWTP Soerendonk purifies domestic sewage water from the nearby villages for 40 years already. With the renovation of this SWTP, Water Board De Dommel started from a broad range of ambitions. Technically the purification performance of the WWTP is improved. The integration into the landscape is guaranteed by care for the architecture, preservation of image-defining green structures and the strengthening of the Ecological Main Structure elements. The hiking and biking trails along the stream are integrated in a regional recreational network. A full-scale water harmonica, in which the effluent of the sand filters, oxygenated and rhythmically treated in the flowform® cascade, is cleared of bacterial remains and is grafted with its characteristic brook flora and fauna, is the integral coping stone of these ambitions. Improvement of effluent quality, water storage in the valley, enhancing the natural, scenic embedding, public contact and recreational and educational goals are all well served.

The water harmonica

The design for the water harmonica Soerendonk is composed of three sections, surrounding the conventional SWTP: flea pond (1), reed swamp (2) and biotope (3). The sand filter makes it possible to save space, because by this filter the load of the suspended matter in flea ponds decreases. In addition, space saving is achieved by using the reed marsh partly as a filter with submerged aquatic plants. The full waste water effluent of 40.000 residents (5000 m³ / d) is treated in the harmonica, the maximum hydraulic capacity is 1000 m³ / h. The time the water stays in the harmonica is 4 days in total.

The terrain of the entire water harmonica is open to the public. A bicycle route leads to a bridge over the flea ponds in section 1 (yellow line, to the right of 1), with a view on the flowform® cascade. Via a footpath the two sections 2 and 3 can be visited and experienced.



Flowform® cascade

Between the sand filter building of the conventional installation (right above 1) and the water flea ponds, a flowform® cascade (right of 1) has been built which will give the effluent a rhythmical treatment.

For the first time in the world the effluent stream of a SWTP will integrally undergo a rhythmic treatment, which not only will increase the amount of oxygen in the water, but which also is expected to have a stimulating effect on the ecology of the harmonica and the brook. The flowform® units have been designed by the artist Paul van Dijk, who faced the challenge to translate the existing small-scale models into a technically feasible concept for large flow rates, and for a very slight slope (5%). The extremely lively water movement can be observed from the nearby bicycle bridge.

The cascade consists of 45 rows of flowform® units next to each other, together creating a plane of about 30 x 6 meters. One row holds 6 units. Each unit consists of 3 flowforms. Consequently such a row holds 18 flowforms.

The total water supply will be distributed equally among these 45 rows. Each row can have between 50 and 140 litres per minute running through. The total of 45 rows allows for a maximum of 380 m³ per hour.



A flowform® unit, a design of Paul van Dijk, measures ca. 50 x 100 x 15 cm, is made of polyester filled with p.u. foam (fully enclosed) and contains 3 flowforms.

Rhythm and Flowform® Technology

Water is the bearer of life. Healthy water is of crucial importance for the emergence and the maintaining of life processes.

Paul van Dijk: "by careful observation of water's behavior in different natural situations, it becomes clear that it is water's nature to react with rhythm on the slightest form of resistance. And in particular with a path of left and right spiraling vortices. Think for example of a meandering river, or the range of vortices generated behind a pole in a river. *With rhythmic spiralling vortices water is 'in its element'.*"



A path of vortices, made by a straight-line movement from left to right with a brush in standing water, made visible by white powder on the surface.

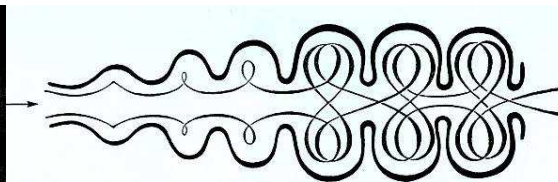


Diagram of the movements of water through a flowform cascade.

Paul: "In designing a flowform® I try to 'structure' the resistance by creating subtle relationships and bends in the surfaces, so that the water reacts with a persistent rhythm in the flow pattern of a lemniscate. This I call the flowform® principle. A flowform® design has been developed empirically according to the nature of water itself. Water that has been rhythmically treated this way, will be able to be bearer of a variety of life processes".



World first

It is for the first time that the flowform® principle finds its application on such a scale, for such amounts of water and with so little gradient (here was very little fall available, only 30cm). This project has been completed in spring 2012.



Literature: J. Schwuchow e.a.: Energizing water, flowform technology and the power of nature. Sophia Books, UK 2010.
John Wilkes: Flowforms. The rhythmic power of water. Floris Books, UK 2003.
Theodor Schwenk: Das sensible chaos. Verlag Freies Geistesleben. Stuttgart (DE) 1962

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